

A Computational Comparison of Prototype, Exemplar, and Rule-Based Models of Human Category Learning

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Abstract—I implement three computational models of concept representation—a prototype model, an exemplar model, and a rule-based model—and evaluate which of them reproduces the human difficulty ordering of the six Shepard, Hovland, and Jenkins (SHJ) category structures. The criterion is behavioral rather than accuracy-based; a model succeeds to the extent that the structures difficult for it are the structures difficult for people, reproducing the ordering $I < II < \{III, IV, V\} < VI$ and, in particular, the comparatively early mastery of the two-dimensional Type II structure. Each model is implemented as an online learner and trained under the errors-to-criterion paradigm, and the exemplar model is realized as ALCOVE so that its selective attention is learned trial by trial rather than fixed in advance. I find that only the exemplar model with learned attention reproduces the ordering, including the Type II advantage; freezing attention removes that advantage while leaving the remainder of the ordering intact, which localizes the Type II effect to dimensional attention rather than to exemplar storage as such. The prototype model cannot separate Types II and VI, whose category means coincide, and the rule model masters Type I almost immediately yet finds Type II hard, inverting the human pattern. These results support an exemplar-plus-attention account of human category learning.

Index Terms—category learning, concept representation, exemplar models, selective attention, Shepard–Hovland–Jenkins, cognitive modeling

I. INTRODUCTION

Concept learning is a problem of representation before it is a problem of statistics. A learner who has seen a few members of a category can classify the next member without a definition, and contemporary machine systems approach that competence only in part [1]. Recent comparisons of language-model embeddings against human categorization data find that machine representations align with human category boundaries at a coarse level and diverge on the finer structure of a concept [2]. The question of which representational format best matches human generalization is therefore current rather than historical; it constrains what a more human-aligned representation would have to be.

This project concerns mental representation, which is the central commitment of the computational and representational understanding of mind that frames the course [3], [4]. Cognitive science offers three durable accounts of what is stored when a person holds a concept, and each is a claim about the mind rather than a choice of algorithm. The prototype

account holds that a category is summarized by an abstracted central tendency that need never have been observed [5], [6]. The exemplar account holds that no summary is stored, and that classification is a judgment of similarity to remembered instances [7], [8]. The rule account holds that a concept is an explicit condition on features, acquired by hypothesis testing [9]. Because the three accounts predict different patterns of generalization, they can be separated by evidence rather than left to agree.

I state the research question so that it can fail: among prototype, exemplar, and rule representations, which one best reproduces the way people generalize categories learned from a handful of examples, measured against the human difficulty ordering of the six SHJ category structures [10]? People learn the six structures in a stable order of difficulty, and that order—recovered under controlled replication [11]—is the behavioral signature that any account of human concepts must reproduce. I treat reproducing the ordering as necessary and treat raw classification accuracy as helpful but not sufficient; a model that classifies accurately while misordering the structures has failed the cognitive claim. The Type II structure is the sharpest part of the test, because people master it earlier than its number of relevant dimensions would predict, and a model that cannot recover that anomaly cannot be said to generalize as people do.

II. MODEL/TOOL DESIGN

The tool is a small Python codebase with three parts: an encoding of the SHJ stimuli, the three models, and a runner that trains and scores each under matched conditions. I used NumPy for the numerical work and Matplotlib for the figures, and the analysis is packaged as a reproducible notebook so that every reported number and figure is regenerated from a single fixed random seed. The design is driven by theory rather than by engineering convenience; the differences among the models are intended to be the differences among the accounts of Section I, and each model shares a common training interface so that the runner treats them identically.

The stimuli are the eight objects that vary on three binary dimensions, partitioned into two categories of four. Up to relabeling of the dimensions and the categories there are exactly six such partitions, and I do not take that fact on

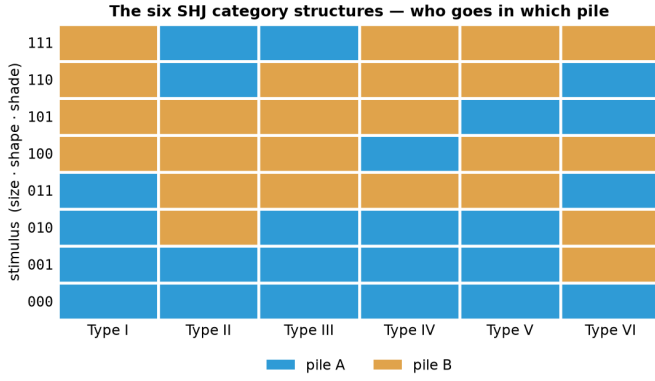


Fig. 1. The six SHJ category structures. Rows are the eight stimuli, given as three-bit codes; columns are the six types; cell color is the assigned category. The partitions are verified by exhaustive enumeration rather than transcribed.

trust; the test suite enumerates every four-against-four split of the stimulus set, groups the splits into symmetry classes, and confirms that the six encoded types match the SHJ structural invariants of relevant dimensions, linear separability, and cohesion (Fig. 1). Encoding the structures by verification rather than by transcription removes a class of silent error that would otherwise be indistinguishable from a modeling result.

The prototype model stores one summary point per category and classifies by similarity to it [5], [6]. Each category’s prototype is the running mean of the instances assigned to it, and the probability of a category is a Luce choice over the similarities of the probe to the two prototypes. This model predicts a specific failure: when the two category means coincide, as they do for the XOR-like Type II and the parity-based Type VI, the prototypes are indistinguishable and the model remains at chance regardless of training.

The exemplar model stores every training instance and classifies a probe by its summed, attention-weighted similarity to those instances, which is the Generalized Context Model directly [7], [8]. The errors-to-criterion paradigm, however, requires trial-by-trial learning from feedback, and a context model whose attention weights are fixed in advance is not a learner in that sense. I therefore realize the exemplar account as ALCOVE, which augments the context model with association and attention weights adjusted by gradient descent after each trial [12]. This is a deliberate commitment rather than a substitution; ALCOVE is the exemplar account made into an online learner, and its attention-learning rate is the single parameter I manipulate. Setting that rate to zero freezes attention at a uniform setting and yields an exemplar learner that can still strengthen its associations but cannot learn which dimensions to weight; setting it positive yields the full model. This contrast is the central manipulation of the study.

The rule model induces an explicit feature-based boundary by hypothesis testing [9]. Its hypothesis space is the set of single-feature rules together with two-feature conjunctions, and it follows a win-stay, lose-shift policy: it retains its current rule while that rule is consistent with the instances seen,

and on an inconsistency it moves to the simplest rule that maximizes consistency with everything seen so far. The model masters a single-feature structure immediately, and it has no hypothesis that separates the XOR-like Type II; consequently it is expected to find Type II hard.

The runner follows the errors-to-criterion paradigm of the standard SHJ replication [11]. A block presents all eight stimuli once in random order; on each trial the model emits a category probability, a response is sampled from that probability, the response is scored against the true label, and the model updates from feedback. Training is bounded by a fixed horizon rather than continued to a learning criterion, because some model-by-structure cells never reach criterion—the prototype model on Types II and VI is the clear case—and an open-ended loop would not terminate there. I report the mean total errors within the horizon, averaged over many simulated subjects with independent random trial orders, as the analogue of human errors to criterion; a model’s failure to reach criterion is recorded as a result rather than treated as a fault.

III. RESULTS

I trained four conditions—the exemplar model with attention, the same model with attention frozen, the prototype model, and the rule model—on all six structures, averaging 800 simulated subjects over 25 blocks at a fixed seed. Only the exemplar model with learned attention reproduces the human ordering $I < II < \{III, IV, V\} < VI$, including the position of Type II below the tied III–V band (Fig. 2). A rank correlation between each model’s errors and the human difficulty rank is highest for this condition ($\rho = 0.83$) and lower for every other condition; the ordinal test is nevertheless the decisive one, because the human ranks are tied for Types III, IV, and V and a correlation cannot express that tie.

The Type II advantage depends on attention specifically. With attention learning enabled, Type II is mastered with fewer errors than the III–V band; with attention frozen and nothing else changed, Type II rises above that band and the advantage disappears, while the remainder of the ordering is largely preserved (Fig. 4). The manipulation therefore localizes the Type II effect to the learning of dimensional attention rather than to exemplar storage as such; storing instances is necessary for the effect but is not by itself sufficient to produce it.

The prototype model fails where its structure predicts. It learns the single-feature Type I readily and the linearly separable Type IV moderately well, but it almost never reaches criterion on Types II and VI, whose category means coincide; criterion is reached on roughly eight and seven percent of subjects, respectively. Its errors are large on the two structures that people find, respectively, comparatively easy and hardest, so it misorders the middle of the range and cannot be reconciled with the human pattern.

The rule model presents the opposite profile. It solves Type I in on the order of one error per subject, which no other model matches, but it finds Type II as hard as the three-dimensional structures and does not reliably reach criterion on them. Its ordering places Type II late rather than early,

Difficulty by structure — does the model err where people do?

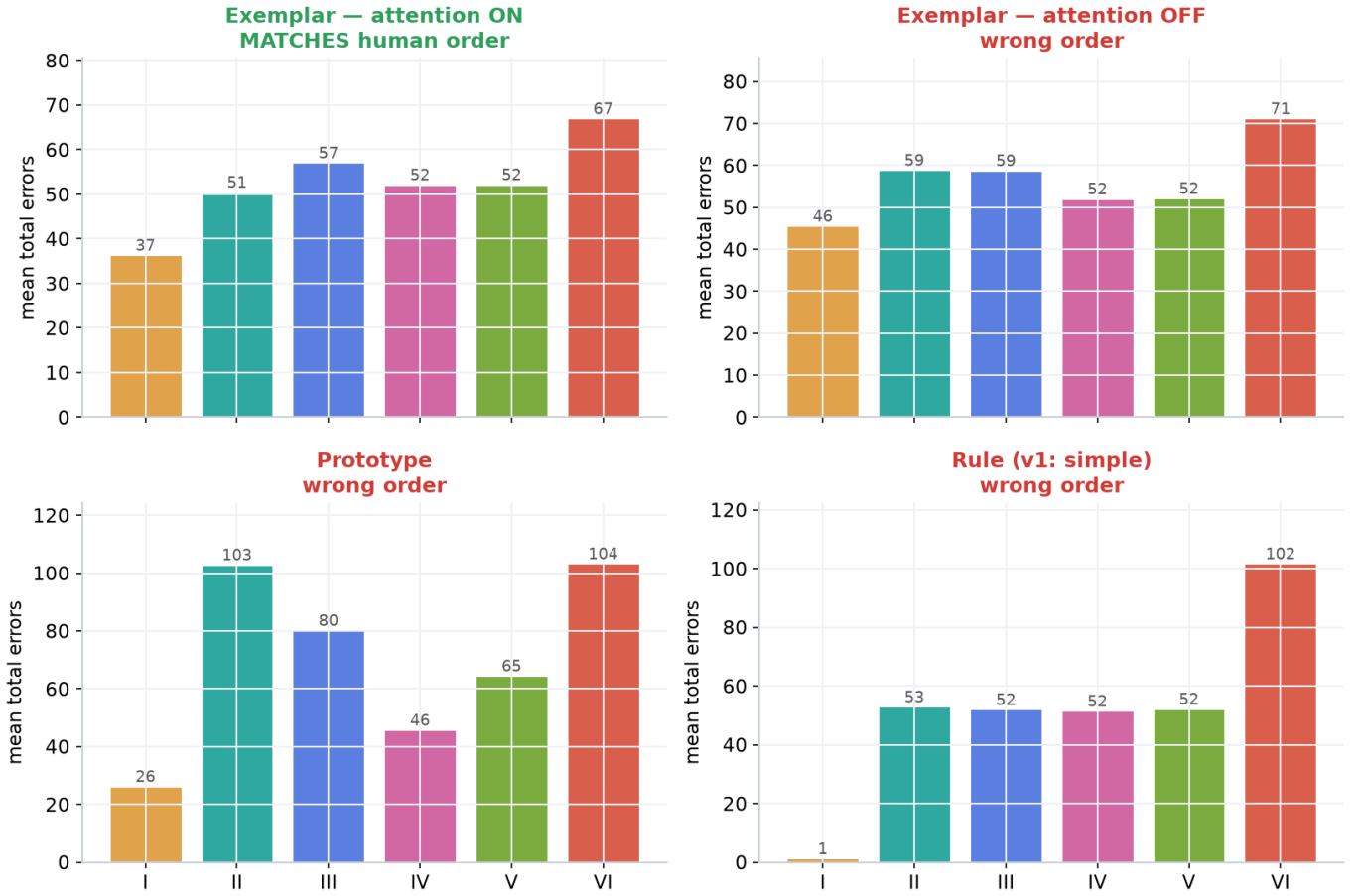


Fig. 2. Mean total errors by structure, one panel per model. A panel is marked as matching the human order when errors satisfy $I < II < \text{band} < VI$, where band is the mean of Types III, IV, and V. Only the exemplar model with attention matches.

which inverts the human ordering at precisely the point that most sharply distinguishes the accounts. The learning curves in Fig. 3 show the same result dynamically: the exemplar-with-attention curves descend on every structure, whereas the prototype curves for Types II and VI do not descend at all.

IV. DISCUSSION

The results support an exemplar-plus-attention account of human category learning. The exemplar model reproduces the human ordering only when it can learn what to attend to, and the ablation shows that this capacity, rather than instance storage in general, is what makes the Type II structure easy. The comparison is informative because the models were built as theories rather than as tuned classifiers; the prototype model’s blindness to Types II and VI and the rule model’s inversion of Type II are consequences of their representational commitments, not of implementation choices, and they are the behaviors that each account predicts in advance.

The finding bears on the alignment of machine representations with human ones. Learned dimensional attention is a

mechanism for ignoring irrelevant structure, and the evidence that machine embeddings match human categories coarsely while diverging on finer structure [2] is consistent with the absence of such a mechanism rather than with a shortage of data. A representation intended to generalize as people do, from few examples [1], would need not only to store instances but to learn which of their dimensions matter; the Type II result is a compact demonstration of why that capacity is necessary.

V. LIMITATIONS

The rule model is the simplest of the three, and its scope limits the strength of the rule comparison. A rule-plus-exception account, in which a learner adopts a simple rule and memorizes the instances that violate it [13], is a fairer model of human rule use and might narrow the gap on Type II; the present rule model therefore establishes a lower bound on what a rule account can do here rather than the strongest such account.

I reproduce the ordinal difficulty ordering rather than the quantitative human learning curves. The model parameters

Learning curves — errors per block over training

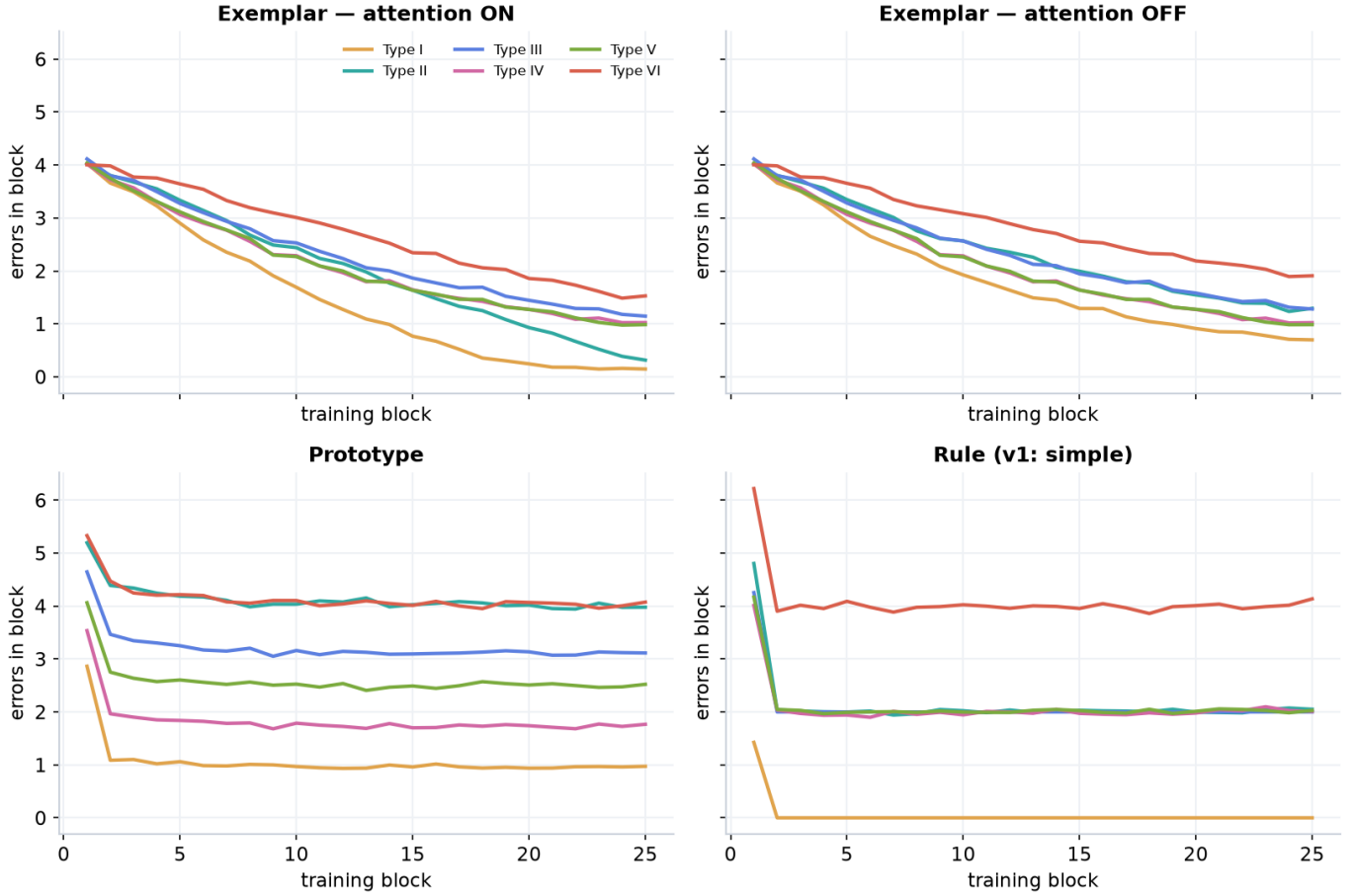


Fig. 3. Learning curves: mean errors per block over 25 training blocks, one panel per model, one line per structure. Descending curves indicate learning; the prototype curves for Types II and VI remain flat.

were set to standard published values and were not fit to human data, so the comparison is a test of qualitative ordering, which is the behavioral signature the benchmark defines, and not a claim of quantitative fit. Fitting the parameters to published learning curves would be a stronger test and is left to future work.

The secondary check that most directly separates the prototype and exemplar accounts, the Medin and Schaffer 5-4 category structure [7], is designed but not yet run, because it uses a four-dimensional stimulus set while the current models are specialized to the three-dimensional SHJ stimuli. The simulated learners are, finally, not a human replication; the human ordering is taken from the established literature [10], [11], and the contribution is a demonstration of which representational account recovers that ordering.

VI. CONCLUSION

I compared prototype, exemplar, and rule models of concept representation on the six SHJ category structures and asked which reproduces the human difficulty ordering. Only the

exemplar model with learned selective attention reproduces the ordering, including the early mastery of Type II, and an attention-freezing ablation localizes that advantage to dimensional attention rather than to exemplar storage in general. The prototype model cannot separate the structures whose category means coincide, and the simple rule model inverts the human position of Type II. Future work extends the rule account to a rule-plus-exception model, adds the Medin and Schaffer 5-4 check with dimension-general implementations of the three models, and fits the model parameters to human learning curves in place of the present ordinal comparison.

ACKNOWLEDGMENT

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Attention is the mechanism behind the Type II advantage

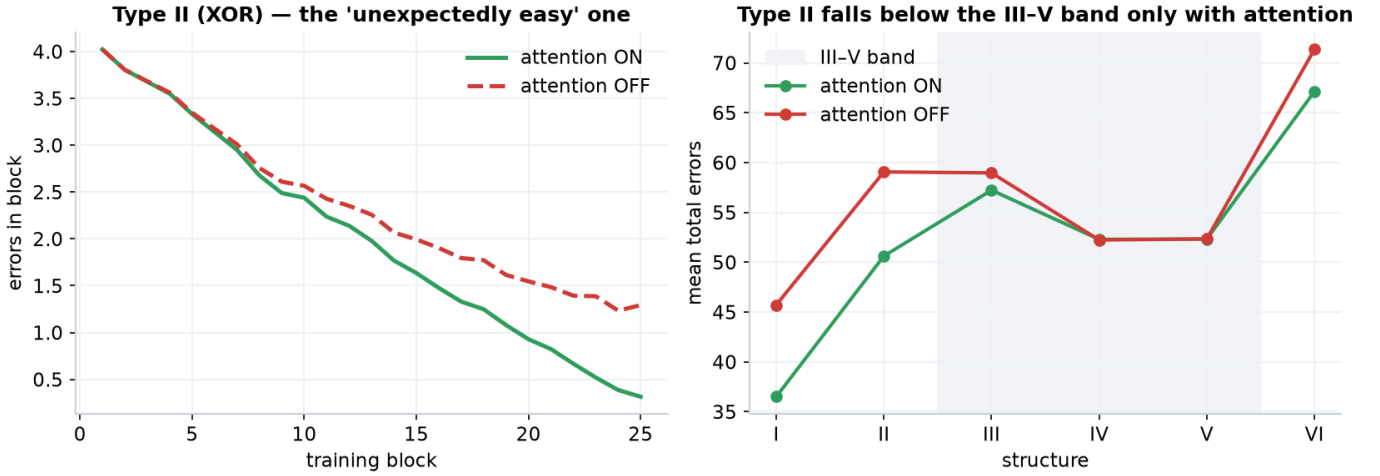


Fig. 4. The attention ablation. Left: the Type II learning curve with attention enabled and frozen. Right: total errors by structure for the two conditions, with the III–V band shaded. With attention frozen, Type II no longer falls below the band.

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APPENDIX A: PROJECT PLAN SUMMARY

The plan below follows the course planning template. Total estimated effort is approximately 70 hours. Milestone deadlines are shown as divider rows. Most implementation tasks are complete; the Medin and Schaffer 5-4 secondary check and the presentation deliverables remain.

APPENDIX B: CODE REPOSITORY

The complete implementation—the stimulus and structure encoding, the three models, the experiment runner, the reproducible analysis notebook, and the test suite that verifies the six category structures by enumeration—is available at the private course repository:

<https://github.com/RabeaWahab/gt-cs6795-concept-learning>

The repository includes a `README.md` with setup and execution instructions; the reported figures and numbers are regenerated by running the notebook `notebooks/model_comparison.ipynb` from a fixed random seed.

Wk	#	Task	Hrs	Done
2	1	Create task list and choose research question	0.5	Y
2	2	Survey concept-learning literature and confirm tool track / scope	3.5	Y
3	3	Collect and verify core scholarly references	1.5	Y
3	4	Write project pitch (all sections) and format in IEEE L ^A T _E X	3.5	Y
<i>Project Pitch Due</i>				
4	5	Set up repository, environment, and project structure	1.5	Y
4	6	Encode SHJ category structures and stimuli	3	Y
4	7	Implement the three core models (prototype, GCM/ALCOVE, rule)	11.5	Y
5	8	Build experiment runner and run models on SHJ structures	5	Y
5	9	Generate comparison figures and draft optional check-in	4	Y
<i>Optional Midpoint Check-in Due</i>				
6	10	Code debugging and architecture refactoring	7	Y
6	11	Add Medin & Schaffer 5-4 secondary check	2.5	N
6	12	Analyze model fit against human difficulty ordering	3	Y
6	13	Expand reference set to 10+ for final report	1.5	Y
7	14	Draft full report (abstract through conclusion)	10	Y
8	15	Revise report, integrate citations, finalize repository	6	In prog.
8	16	Update project plan and submit final report	1.5	In prog.
<i>Final Report Due</i>				
9	17	Build poster and write/rehearse 5-minute script	4	N
9	18	Record video, assemble submission PDF, verify and submit	3.5	N
Total Estimated Hours			73	